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QEEG coherence patterns related to mathematics ability in children

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ABSTRACT

The current study investigates the utility of resting-state EEG coherence values in predicting standardized math scores in children. Quantitative EEG and standardized academic achievement analyses were performed on 60 school-aged children. Analyses assessing intrahemispheric coherence at rest were conducted across the sample of participants and several coherence networks were extracted and compared to standardized math achievement values. Specifically, networks that included Brodmann area 40 (a brain region involved in the cognitive processes responsible for mathematics performance) and whose coherence values were significantly correlated with standardized math scores were examined. Results indicate a total of four coherence networks, two in each hemisphere, that have utility in predicting general math skills in children. In addition to BA 40, these coherence networks include BAs in the right temporal lobe, right frontoparietal lobe, left superior temporal lobe, and the left medial prefrontal cortex. These findings address the current dearth of research on the neurological connectivity patterns that are foundational for mathematics abilities in children. Further, these results lay a foundation for the supplementary use of EEG in the assessment and identification practices surrounding math learning disabilities in children and additionally provide a neurocognitive framework upon which intervention research may be targeted.

KEYWORDS

Assessment; EEG; learning disability; mathematics; QEEG

Introduction

Mathematical reasoning is one of the most vital cognitive skills that individuals learn in early childhood (Rivera et al., 2005). Individuals who experience difficulties acquiring mathematics skills in childhood are at greater risk for academic failure, development of a lifelong mathematics learning disability, and reduced economic success into adulthood (Ritchie & Bates, 2013; Garnett, 1998; Johnson & Blalock, 1987). Often, mathematics deficits develop in the early part of the math learning sequence with core skills including counting knowledge, counting speed, and working memory (Geary, 1993). Given the increased likelihood of comorbid deficits for those with mathematics disabilities, it is crucial for clinicians to have the capacity to adequately and efficiently identify children who may need early math interventions. Despite recent improvements to the procedures that have been put in place by school systems surrounding the identification of students with specific learning disabilities (SLDs), there is still much room for growth and improvement.

It is estimated that 3–8% of students suffer from a form of math disability worldwide, though estimates vary depending on how researchers operationalize and define the disorder (Van Luit & Toll, 2018). The current methods used to identify children with learning disabilities (LDs), as guided by U.S. federal and state legislation, have significant problems (Decker et al., 2012; Fletcher et al., 2011). Identification approaches that are currently relied upon, such as the IQ-Discrepancy Model—colloquially referred to as a "wait-to-fail" approach—are often only deployed for children after they have fallen far behind in a school subject area; resulting in years of intervention to merely catch up to peers. The alternative approach, Response to Intervention (RTI), which was designed to address this problem and improve early identification, does not lead to improved outcomes (Balu et al., 2015). Additionally, despite the perceived propensity for early intervention that the RTI method offers, failure to respond to an intervention does not provide sufficient diagnostic information to identify a disability, thus highlighting another major flaw in this approach. As such, the potential for supplementary and/or more reliable methods to identify children with LDs within the school systems, including math LDs, exists and should therefore be pursued.

A likely reason for the issues seen regarding improper diagnosis and intervention for children with specific learning disabilities can be explained by the relatively large gap in the neurocognitive research literature on these topics. Although there have been advancements in the use of neurological markers to identify those with and without learning disabilities (Alahmadi, 2015), no research has investigated patterns of brain connectivity to empirically predict mathematics ability in children. The current study aims to explore children's basal neuro-electrophysiological activity to

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determine if brain coherence at rest is predictive of general math skills. This information will help to shed light on the underlying neurocognitive mechanisms fundamental to mathematical ability and thus, guide future research on performance-based math assessment batteries as well as math interventions for struggling children.

Electroencephalography (EEG) approaches are one such method to investigate a person's neurocognitive processes that underly specific behaviors and abilities since these approaches have the potential to highlight the neurological underpinnings of cognition. Additionally, findings from EEG research help to inform specific interventions and therapeutic applications such as neurofeedback therapy. Consequently, there is a growing interest for the use EEG in both the research and practical domains of healthcare, as this imaging modality offers up the potential for growth in many spheres of psychological clinical practice. One such EEG analysis technique involves the investigation of coherence, a type of quantitative EEG (qEEG) analysis that is often used when examining continuous EEG data (González-Garrido et al., 2018; John et al., 1988; Thatcher et al., 1989, 2001, 2005). Specifically, EEG coherence refers to the degree of brain activity synchrony between specified locations throughout the brain (Gasser et al., 1987); it can therefore be used to determine which brain networks are "in sync" with one another during the EEG recording and which general brain regions are involved in those particular networks (Dumermuth, 1973; González-Garrido et al., 2018).

One exploratory research technique involves the investigation of an individual's coherence while they are not performing a task. A coherence recording at rest (subject not performing a task) implies a recording of the individual's basal electrophysiological brain functions, which has been termed the "EEG-Default Mode Network" by Chen et al. (2008). Examining an individual's default EEG brain activity can provide information about their neurocognitive aptitudes or, in some cases, incapacities (Margolis et al., 2019; Rocca et al., 2010; Savini et al., 2019).

Neuroimaging and math

In 1992, Dehaene proposed a theoretical neurocognitive model to outline mental arithmetic and numerical processing entitled "the triple-code model." Three years later—with the advent of fMRI research and its corresponding studies thoroughly underway—Dehaene and Cohen (1995) expanded upon this model by reviewing the available fMRI case studies exploring this topic. Through their review of the relevant fMRI research, the triple-code model was expanded to create a cohesive functional-anatomical model of number and arithmetic mental processing. This updated model hypothesized the anatomical correlates of mathematical ability and number processing. Specifically, the updated triple-code model posits that numbers are processed by three distinct brain areas: (1) visual number processing occurs in the ventral occipito-temporal areas, (2) quantity and magnitude judgements occur in the inferior parietal areas (Brodmann area 40), and (3) the mathematical verbal code is processed in the left perisylvian areas in the inferior parietal lobules. The model has since been empirically validated (Schmithorst & Brown, 2004) and further expanded upon with the availability of additional fMRI data (Arsalidou & Taylor, 2011).

Arsalidou and Taylor (2011) updated Dehaene and Cohen (1995) model by reviewing the extant fMRI research exploring cognition as it relates to math ability and creating an atlas of the functional anatomical correlates for mental arithmetic operations. The atlas indicates that both the left and right inferior parietal lobules (Brodmann area 40: BA 40) are the most likely brain regions to show activation during general numerical processing and calculation tasks. Additionally, their review revealed that BA 40 showed a relatively high likelihood of activation during addition tasks, a moderate activation likelihood during subtraction tasks, and a relatively low likelihood for activation during multiplication tasks (Arsalidou & Taylor, 2011). These findings suggest that the cognitive processes specialized to BA 40 play a crucial role in a child's basic processing of numbers and performance of simple mental calculations; a finding that further supports the theories first presented in Dehaene's Dehaene (1992) functional-anatomical triple-code model.

Quantitative EEG has a history of being used to assess underlying brain functions for various neuropsychological disorders (e.g., Thatcher et al., 1989, 2001). Numerous studies have also utilized qEEG to examine its relationship with measures of intelligence, several of them reporting significant relationships between coherence and standardized intelligence measures (Martín-loeches et al., 2001; Neubauer & Fink, 2009; Thatcher et al., 2005). Additional studies have cemented the utility of qEEG to study neuropsychological differences in individuals with ADHD, demonstrating its utility in this domain with several decades worth of literature (Barry et al., 2009; Clarke et al., 1998, 2001; Janzen et al., 1995; Satterfield et al., 1972).

Despite an extensive history of utilizing qEEG for studying various neurocognitive phenomena, a relatively small base of literature exists outlining its utility for examining children's academic skills and abilities. While recent EEG studies have begun to focus on children's reading abilities and disorders (e.g., dyslexia; Bosch-Bayard et al., 2020), very few EEG studies have examined the underlying neurocognitive mechanisms involved in general mathematics skills.

Although arithmetic abilities have been studied for many years within the frame of an educational context, neurocognitive research on mathematics abilities is a relatively recent field of study (Plerou & Vlamos, 2016). One of the most recent research studies on this topic by González-Garrido et al. (2018) explores mathematical achievement as it relates to qEEG measures of coherence. Specifically, this study aimed to evaluate the coherence levels in children with differing math skill abilities as they performed math tasks and examine coherence differences between children who performed particularly well in math tasks against those who performed poorly. Findings from this study suggest that there are no specific math brain networks in children and further posit that a child's mathematical abilities rely upon a

complex integration of several connected brain regions and networks (González-Garrido et al., 2018). This theory is in support of findings from prior research depicting EEG differences during mental calculation tasks (Fernández et al., 1995). Additionally, González-Garrido et al. (2018) posit that higher math achieving children display more localized coherence over parietal areas—particularly in the beta band—than low achieving children, which is in alignment with prior fMRI research (e.g., Zhang et al., 2017). Thus, we expect children with greater coherence around BA 40 (a BA in the parietal brain region) to display greater math abilities than those with lower coherence values around this region.

A significant number of studies have implicated Brodmann area 40, synonymously referred to as the inferior parietal lobule (IPL), as having an important connection with math ability (Anderson et al., 2011; Cohen et al., 2000; Kroger et al., 2008), though many of its theorized functions remain contested. Experts propose that this brain region plays crucial roles in spatial perception, visual motor integration (Andersen, 2011), tactile perception, manual construction (Jäncke et al., 2001), working memory (Baldo & Dronkers, 2006), decision making (Vickery & Jiang, 2009), sustained attention (Singh-Curry & Husain, 2009), and mental mathematical operations (Anderson et al., 2011; Cohen et al., 2000; Kroger et al., 2008).

Several techniques have been used to study the functionality of this brain region. Lesion studies indicate that patients who have suffered damage to the IPL often suffer from multiple syndromes including aphasia, dyslexia, visual-spatial neglect, and dyscalculia (Cohen et al., 2000; Singh-Curry & Husain, 2009). Neuroimaging research on this region further supports the results from lesion studies, showing that the IPL plays a critical role in language and calculation abilities (Cohen et al., 2000; Rivera et al., 2005). Interestingly, further findings show that children who are more adept in mathematics have greater activation in the IPL with less activation in other regions, and therefore conclude that this brain region has a functional specialization for mental arithmetic processes (Rivera et al., 2005).

Current study

Although progress has been made in improving our understanding of the neurocognitive mechanisms underlying mathematical abilities (e.g., Arsalidou & Taylor, 2011; González-Garrido et al., 2018), experts in the field still struggle to translate this basic neurocognitive research into practical outcomes (Plerou & Vlamos, 2016). Recent translational research has begun investigating the utility of neurophysiological resting-state paradigms to evaluate biomarkers of various clinical and neurological disorders (Takamura & Hanakawa, 2017). The current study utilizes a qEEG resting-state paradigm to explore children's basal electrophysiological brain activity to determine if coherence values among brain regions at rest are predictive of general math skills. Although task-based EEG and fMRI research on math disabilities elucidates key brain areas actively involved in completing specific math tasks, there are several advantages to resting-state paradigms, which were taken advantage of in the current study to uncover associations between brain networks and later math performance.

By using a resting-state paradigm, we gathered continuous sets of EEG data from child participants and extracted coherence values between BA 40 in each hemisphere and their connectivity with the rest of their respective hemispheres. Using EEG resting-state coherence to investigate BAs that are significantly coherent with BA 40 allowed for the identification of specific basal electrophysiological brain activity that is linked to standardized math achievement scores, and therefore, general sets of math abilities (Anderson et al., 2011; Cohen et al., 2000; Kroger et al., 2008).

Table 1. Demographic characteristics.

Participants	
N	60
Gender (%)	
Male	53.3
Female	46.7
Age (years)	
Mean	9.58
SD	1.38
Ethnicity (%)	
Caucasian-Nonhispanic	83.3
African American-Nonhispanic	8.3
Latino, Hispanic, Spanish Origin	1.7
Asian, South-East Asian	6.7

Methods

Participants

Data used in the current study was derived from a larger parent study aimed at examining the relationships between brain function, math performance, and anxiety. Participants for the current study included 60 school-aged children (ages 7–12 years), including typically developing children, children with low math achievement (LA), and children with a suspected and/or confirmed math learning disability (MLD). This sample was utilized to ensure the full range of mathematical achievement scores was obtained for the study sample. Children were excluded from the study if they were deemed to have an intellectual disability, as determined by their Broad Cognitive Ability score from the Woodcock-Johnson Tests of Cognitive Ability—3rd Edition (WJ-III Cog; Woodcock et al., 2001) falling below the score of 70. Demographic characteristics of the sample are reported in Table 1.

Measures

The current study used the WJ-III Ach to determine mathematical abilities. This battery is designed to measure an individual's academic skills whom are aged two to 90 or more years, and it has been validated for its reliability and consistency in research studies. Its core subtests have median reliability coefficients of r_{11} = .81–.94 (Note, the Broad Cognitive Ability score on the WJ-III Cog was used as an estimate of IQ, to rule-out general cognitive

impairment for children scoring low on math measures used in the inclusion criteria). The three core math subtests were used for the current study and include the Math Calculation, Applied Problems, and Math Fluency subtests. The Math Calculation subtest requires students to write down the answers to increasingly difficult mathematics problems (ranging from counting problems to algebra). The Applied Problems subtest includes math problems verbally read to the student by the examiner. Students answer verbally on the Applied Problems subtest but are allowed a paper and pencil if needed. Lastly, the Math Fluency subtest requires students to answer as many simple math problems as they can in a specified timeframe.

EEG data were collected to examine whether specific frequency bands (i.e., delta, theta, alpha 1, alpha 2, beta 1, beta 2, beta 3, and high-beta) and the coherence patterns among them were able to predict the degree of a child's mathematical abilities. EEG data were recorded from 19 channel electrodes distributed across the scalp via Electro-cap using the 10/20 placement system. Data were sampled at 1026 Hz using a BrainMaster Discovery 24E amplifier. This device was selected due to its FDA approval classification as well as its compatibility with the Neuroguide program 6.6.4 (Thatcher, 2011). A 60 Hz notch filter was used to remove noise caused by electronics from the surrounding environment and the bandwidth range was set to record frequencies between 1.0 and 30 Hz. The frequency bands used in the current study are defined as follows: delta (1.0–4.0 Hz), theta (4.0–8.0 Hz), alpha 1 (8.0–10.0 Hz), alpha 2 (10.0–12.0 Hz), beta 1 (12.0–15.0 Hz), beta 2 (15.0–18.0 Hz), beta 3 (18.0–25.0 Hz), and high-beta (25.0–30.0 Hz). Impedance values for the A1 and A2 ear reference electrodes were kept below 5KΩ, and all other electrode impedance values were kept below 10KΩ for all subjects. Neuroguide 6.6.4 (Thatcher, 2011) was used for removing EEG artifact in the data and to obtain normative values of qEEG spectral coherence and MATLAB (2018) was used for data transformation and organization.

Procedures

Prior to conducting the study, approval to perform the research procedures was granted from the University of South Carolina's institutional review board. Participants were provided child assent and parental consent forms and signatures were obtained. Preliminary measures of mathematical skills and cognitive abilities were obtained from participants who agreed to partake in the study. Data from participants who met the study eligibility criteria were retained and EEG data were recorded. EEG recordings were obtained by fitting the participants with their appropriately sized Electro-Cap and ground leads. The recordings were collected over three-minute intervals while the participants were awake, at rest, with their eyes were closed. This method of collecting resting-state data was used due to the advantages it offers for young participants, specifically, three minutes is a relatively short amount of time for a child to remain still and calm, but still allows for the adequate collection of qEEG data to for coherence analyses. Furthermore, having participants close their eyes does not significantly impact brain activity in brain regions unrelated to visual processing (e.g., math regions), and having eyes closed during an EEG recording provides a method to reduce common EEG artifact associated with eye muscle movements (Barry et al., 2007). All data used for the current study were collected over the course of one to two study sessions dependent on participant availability.

Data analysis

Prior to conducting analyses, the first minute of each participant's qEEG data was manually inspected to identify a minimum of ten seconds of artifact-free data. The visual inspection process involves a review of the raw EEG data, where the reviewer aims to identify points in time with abnormal spikes in amplitude across multiple channels at the same time. Artifacts of this type are typically related to scalp and/or facial muscle movement and are therefore removed from the dataset individually for each participant. Following visual inspection, an automated program within the Neuroguide software was used to identify and flag EEG patterns consistent with artifacts related to drowsiness (e.g., slow rolling eye movements, specific changes in alpha rhythm, specific changes in theta and beta ratios, etc.) and eye muscle movement. These flagged patterns were subsequently reviewed for accuracy and manually removed by the reviewer if necessary. The Neuroguide software then uses the artifact-free data from the now "cleaned" 10 s sample as a reference. With this artifact-free reference, the automated software program identifies and selects artifact-free data from the rest of the three-minute dataset and discards all portions of the data with artifacts; thus, yielding an entirely artifact-free EEG sample. This procedure was repeated for each participant. By using the automated artifact flagging and rejections programs housed within the software several unique advantages are offered compared to other methods. These methods are reliant upon objective quantitative values within each participant's dataset, as such, this more objective method of data cleaning removes the potential for confounding inconsistencies when rejecting EEG artifact for each participant.

Coherence measures between electrodes were obtained through an automated Neuroguide program. Neuroguide software contains a proprietary database with information from 625 individuals, covering the age range of two months to 82.6 years (Johnstone & Gunkelman, 2003, pp. 42–43). By sourcing this database, Neuroguide yields reports which provide coherence values in raw Z-score units. Utilizing standardized coherence values allows for the reduction of discrepancies in coherence due to age-related/developmental differences. A subsequent automated Neuroguide program, utilizing Low Resolution Electromagnetic Tomography (LORETA), was performed to convert the standardized coherence values between electrodes into a format that yields standardized coherence values between each of the 52 Brodmann areas in either hemisphere.

LORETA is one of the most extensively used algorithms for localizing the source of EEG signal detected on the scalp (Grech et al., 2008). It accomplishes this by attributing electrode activity to specific regions under the scalp, indexing those regions through a standardized MRI atlas, and assigning BAs to the regions based on the MRI atlas. By running the EEG dataset through the LORETA program 3-dimensional statistical maps were generated to model the distribution of brain coherence values. The obtained values of coherence between scalp electrodes were converted into coherence values between each of the 52 BAs within their respective hemispheres. By obtaining models of EEG activity based on an MRI atlas, the current study utilized the findings by Arsalidou and Taylor (2011), which suggest that BA 40 is heavily implicated in mathematical cognitive processes, to provide a framework for which the analyses would be based off. As such, MATLAB (2018) was utilized to extract intrahemispheric coherence data between BA 40 and all other BAs from each respective brain hemisphere. This data was then exported to Microsoft Excel and IBM SPSS (version 24; IBM SPSS Statistics for Windows, 2017) for final analyses.

Using IBM SPSS (*IBM SPSS Statistics for Windows*, 2017), principle component analyses (PCA) with varimax rotation were applied individually to coherence values across each frequency band of interest (delta, theta, alpha 1, alpha 2, beta 1, beta 2, beta 3, and high-beta) for BA 40 in the left hemisphere then separately for each frequency for BA 40 in the right hemisphere to the rest of the BAs in their respective hemispheres. PCA was used to reduce the number of EEG coherence variables and extract the essential features from the data, thus facilitating a more accurate interpretation of the coherence properties between brain regions in either hemisphere. PCA is a traditional method that has been used in EEG analysis due to the high number of variables that are acquired and the necessity to reduce that number for studies with smaller sample sizes. PCA has been used in previous research to achieve similar analytic goals (Vigário et al., 2000). Only components whose Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy was at or above the recommended value (KMO = .60; Dziuban & Shirkey, 1974), had eigenvalues greater than 1.0 (Gorsuch, 1990), and that passed the scree test (Bro & Smilde, 2014) were considered for ensuing analyses.

PCA revealed that BA 40 was involved in components across various frequencies in both hemispheres. Subsequent bivariate Pearson's correlation coefficients were computed to quantify associations between EEG coherence components and math performance The components whose correlations were significant with at least two of the three WJ-III Ach math subtests (p's $< .05$) were used in subsequent multiple regression analyses to assess their utility in predicting each math subtest scores individually. The large number of correlation tests run and the relatively small sample size (i.e., 45 components correlated with each of the three math subtests, yielding 135 correlations on 60 participants) introduced the issue of random false positive results due to the high likelihood family-wise type 1 errors. To minimize the chances of Type I error across such a large number of tests, we required a coherence component to correlate with at least two out of the three math subtests to meet criteria for inclusion in the subsequent regression analyses. These criteria assumed it is unlikely that a component would correlate with two (or all three) math subtests by chance and enabled us to focus on brain activity related to broad, rather than specific, math skills. Thus, despite not correcting for multiple comparisons in a traditional fashion (e.g., Bonferroni correction), this strategy does account for and effectively reduce the occurrence of random false positive results. The last step of the analyses used each of the significantly correlated coherence components to predict both broad math ablity and broad reading ability scores to determine if the extracted coherence components were specifically associated with math, or academic ability more generally (i.e., discriminant validity).

Table 2. Sample descriptive statistics for WJ-III Ach Math Standard Scores.

Subtest	*M*	*SD*	Min/Max
Calculation	101.35	20.30	52–148
Math fluency	103.33	17.60	60–145
Applied problems	93.50	16.30	67–134

Results

Descriptive statistics for all participants' scores on the WJ-III Ach math subtests including means, standard deviations, minimum scores, and maximum scores are reported in Table 2. Examination of the descriptive statistics indicates that the current study included a sample of participants whose math achievement scores are generally representative of the range of scores observed in the population.

Eligible components (eigenvalues greater than 1.0, passed the scree test, had a KMO value greater that .60) were extracted separately for each frequency band in both brain hemispheres using varimax rotation. Table 3 lists all components that were extracted meeting these criteria for each wavelength by hemisphere with the percent of variance explained by the Eigenvalue of that component.

Associations with broad math ability

Pearson's correlations were used to examine the linear relationships between the extracted EEG components (Table 3) and the WJ-III Ach math scores. A total of three coherence components were significantly correlated with all three math subtest scores and one component that was correlated with two out of the three scores. It was assumed that if a component correlated with at least two out of three math subtests it was indicative of brain activity related to broad, rather than specific, math skills. The four extracted components indicative of broad math abilities included: right hemisphere alpha 1 component one, right hemisphere theta component four, left hemisphere delta component one, and left hemisphere theta component one.

In order to localize each component to specific regions in the brain, the components were inspected to determine what

Table 3. Extracted components and Eigenvalue variance explained.

Left		Right	
Wavelength	Variance explained	Wavelength	Variance explained
Alpha 1		*Alpha 1*	
Component 1	64%	Component 1	59%
Component 2	10%	Component 2	13%
Component 3	8%	Component 3	9%
Alpha 2		*Alpha 2*	
Component 1	59%	Component 1	54%
Component 2	13%	Component 2	18%
Component 3	9%		
Beta 1		*Beta 1*	
Component 1	24%	Component 1	29%
Component 2	16%	Component 2	15%
Component 3	13%	Component 3	12%
Beta 2		*Beta 2*	
Component 1	36%	Component 1	28%
Component 2	14%	Component 2	17%
Component 3	10%	Component 3	14%
Beta 3		*Beta 3*	
Component 1	37%	Component 1	43%
Component 2	16%	Component 2	17%
Component 3	14%	Component 3	14%
High-Beta		*High-Beta*	
Component 1	42%	Component 1	44%
Component 2	15%	Component 2	13%
Component 3	12%		
Theta		*Theta*	
Component 1	34%	Component 1	31%
Component 2	13%	Component 2	16%
Component 3	11%	Component 3	14%
Component 4	9%		
Delta		*Delta*	
Component 1	37%	Component 1	47%
Component 2	14%	Component 2	14%

individual variables were loading into each. Each variable represents a coherence value between a specified BA and BA 40. Not surprisingly, the BAs represented in by each variable tended to cluster within specific lobes and/or areas of the brain for each component. The variable clusters that comprised each component were manually recorded, and a filter was overlaid on a standard MRI brain template to characterize the brain regions represented by each cluster of variables for each component. Figure 1 provides a graphical representation for the location of the coherence components in their respective frequency bands and hemispheres.

Table 4 depicts correlations between the identified coherence components and each of the WJ-Ach math subtests. In the right hemisphere there are significant associations between brain coherence levels and all three WJ-III Ach math subtests for two components: the first component in alpha 1 (A) and the fourth component in theta (B). The first alpha 1 component is positively correlated with all three

Table 4. Significant Pearson's correlations between PCA components math abilities.

Coherence component	Calculation	Math fluency	Applied problems
Alpha 1, RH, Component 1	.33 (.01)**	.31 (.015)*	.29 (.025)*
Theta, RH, Component 4	−.34 (.007)**	−.41 (.001)**	−.43 (.001)**
Delta, LH, Component 1	.35 (.007)**	.36 (.005)**	.37 (.004)**
Theta, LH, Component 1	.29 (.025)*	.35 (.006)**	.22 (.094)

*$p < .05$.
**$p < .01$.

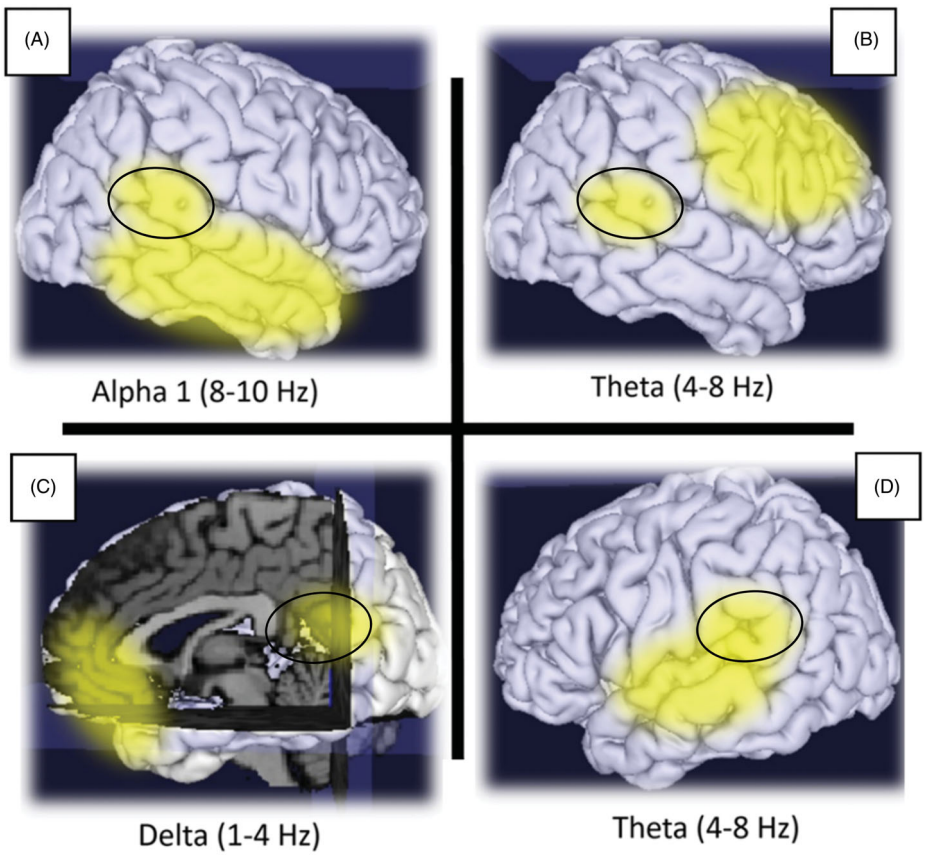

Figure 1. Associated coherence components. (A) Alpha 1 component 1, right hemisphere; (B) theta component 4, right hemisphere; (C) delta component 1, left hemisphere; (D) theta component 1, left hemisphere. *Note*: encircled regions represent the cortical location of Brodmann area 40 (i.e., the surface of the inferior parietal lobe) within the specified coherence component.

Table 5. Multiple linear regression model using coherence components to Broad Math composite score.

	β	F	t	p
*Broad Math Model***		5.88		= .001
Alpha 1, RH, Component 1	.211		1.71	.092
Theta, RH, Component 4	−.251		−1.95	.057
Delta, LH, Component 1	.265		2.13	.037*
Theta, LH, Component 1	.060		0.48	.633

*$p < .05$.
**$p < .01$.

Table 6. Significant Pearson's correlations between math coherence components and Broad Reading scores.

Coherence component	Broad Reading
Alpha 1, RH, Component 1 (A)	.40 (.002)**
Theta, RH, Component 4 (B)	−.36 (.005)**
Delta, LH, Component 1 (C)	.30 (.02)*
Theta, LH, Component 1 (D)	.25 (.06)

*$p < .05$.
**$p < .01$.

math subtests (p's $< .05$). Notably, the fourth theta component is negatively correlated with all three math subtests (p's $< .01$), suggesting an increase in this coherence network is associated with reduced mathematics performance. In the left hemisphere, there are significant associations between coherence and math abilities for an additional two components: the first component in delta (C) and the first component in theta (D). Both coherence components are positively correlated with all math scores. The first delta component (C) is significantly associated with all three math subtests (p's $< .01$), while the first theta component (D) is significantly associated only with the Calculation and Math Fluency subtests (p's $< .05$) but not Applied Problems.

Multiple regression

A multiple regression model including all four components was used to characterize their utility for predicting children's broad math abilities (Table 5). The model explains 30% of the variance and significantly predicts the Broad Math WJ-ACH score, which is a composite of all three math subtests (F (4, 55) = 5.88, $p < .01$, $R^2 = 0.33$). Variance inflation factors (VIFs) for each variable are below two, suggesting low multicollinearity within the model.

Discriminant validity

The predictive utility of the identified components was examined for the broad reading composite score to identify whether these coherence circuits were related to math-specific abilities, or to general academic abilities (Table 6). The right first alpha 1 component, the left first and right fourth theta components, and the left first delta component were each examined to determine their correlations with broad reading ability using broad cluster scores provided by the WJ-III Ach.

Three of the four components, (excluding the left first theta component (D)) are correlated to reading ability ($p > .05$). The components are positively related to math and reading ability, except for the right fourth theta component (B), which is negatively related to both. Overall, these results suggest that, despite including BA 40, components A, B, and C may be tied to general academic ability and/or neurocognitive systems that are foundational to academic performance, rather than mathematics processing specifically. Notably, the left first theta (D) component is related exclusively to math scores, and not reading scores, suggesting a domain-specific component for mathematics.

Further, a multiple regression model including all four components was examined to determine if the model significantly predicts the Broad Reading score. The model explains 26% of the variance and does, in fact, significantly predict Broad Reading scores, F (4,55) = 4.79, $p < .01$, indicating a domain general model. Interestingly, if the right alpha component (A) is removed from the model (predicting Broad Math and Broad Reading), then the model significantly predicts Broad Math scores (F (3,56) = 6.63, p =.001, R^2 = 0.26), but not Broad Reading, (F (3,56) = 3.88, p >.01, R^2 = 0.17). This suggests that the alpha component (A) may be related to domain general cognitive processes, but once that component is removed, the model is more indicative of domain specific cognitive processes for mathematics.

Discussion

The present study investigated BA 40 and its qEEG coherence patterns in school-aged children with a wide range of math abilities. Currently, there is a dearth of research examining neurocognitive underpinnings of brain functions that are implicated in mathematical abilities. This is the first study to examine associations of BA 40 qEEG coherence and math performance. Importantly, the current study adds to cognition literature on mathematics performance by providing empirical support for a unique method to examine brain activity as it relates to academic performance. This method provides evidence of specific brain activity patterns that are implicated in math ability. QEEG measures, such as coherence, provide bio-signatures associated with a child's aptitude for math, and these bio-signatures can be used to enhance the theoretical understanding of the cognitive processes that underly MLD. Additionally, the accumulation of findings from studies utilizing resting-state qEEG paradigms to assess academic ability can one day provide practitioners with the necessary information to improve SLD identification practices and guide intervention techniques.

As mathematics skills are important for a child's future academic and vocational success (Garnett, 1998; Ritchie & Bates, 2013), improving the ability to quickly and accurately determine a child's aptitude for mathematics is particularly important. Improving the theoretical understanding of how a child's brain performs math tasks can be achieved through the examination of neuro-electrophysiological activity that is correlated with math performance. Understanding the underlying neural processes is a crucial step toward addressing the need for improved mathematics assessment and intervention practices.

By introducing a novel method for assessing academic ability, the current study provides foundations for future

basic research surrounding the underlying neurocognitive processes implicated in children's broad mathematical abilities. Despite the utility of more common neuroimaging methods for studying brain activity related to math performance (e.g., event-related potential or functional magnetic resonance imaging research), qEEG resting-state research paradigms offer several advantages that are particularly useful in the context of the current study's aims and goals.

As noted in fMRI research, collecting resting-state data is easier than task-related data, particularly for clinical populations (Heine et al., 2012). Additionally, resting-state qEEG paradigms eliminate researcher bias in picking an appropriate math task to examine broad mathematics abilities. Lastly, using resting-state data allows for examination of connectivity circuits broadly related to math abilities, rather than observing activity only related to the demands of specific mathematics tasks. As such, studies that use resting-state paradigms are more generalizable and comparable to each other given that there is no reliance on stimuli or having to control for behavior.

Research has long supported the notion that BA 40 (IPL) plays a major role in the cognitive processes surrounding math performance (Anderson et al., 2011; Cohen et al., 2000; Kroger et al., 2008), the current study adds to this knowledge-base by elucidating the associations between children's math abilities and intrahemispheric brain connectivity levels with BA 40. Results of the present study indicate that there are two significant coherence networks in each hemisphere that are associated with broad math ability as determined by standardized WJ-III Ach math test scores (see Figure 1).

In the right hemisphere a coherence network in the alpha 1 frequency band (8–10 Hz) significantly predicts standard scores across the Calculation, Applied Problems, and Math Fluency subtests, thus indicating its involvement in mathematics more globally. This network is comprised of BA 40 and various BAs situated across the temporal lobe. The alpha frequency is generally associated with a relaxed state (Milner, 1968) while the function of the right temporal lobe is generally associated with the organization and memory encoding of sensory input (Zulkurnaini et al., 2012), thus this component may highlight the importance of a child's ability to suppress or "relax" the functions of the right temporal lobe in order to perform well in math tasks. Children who exhibit higher levels of suppression, may have a propensity for greater math capabilities due to their ability to block out competing demands on cognitive processes. Similarly, in the right hemisphere, an additional coherence network in the theta frequency band (4–8 Hz) was predictive of global math skills. This network includes BA 40 and various BAs situated in the frontoparietal lobe. The theta frequency band is associated with concentration, where too much can result in impulsivity and difficulty attending (Milner, 1968); the function of the frontoparietal lobes is generally associated with executive control and problem solving (Stuss et al., 1985). Thus, a child's reduced in theta activity in this region compared to other children likely indicates an improved ability to attend and therefore, exhibit the necessary levels of concentration to solve math problems.

In the left hemisphere, a coherence network in the delta frequency band (1–4 Hz) predictive of global math skills included BA 40 and various BAs in the medial prefrontal cortex. The delta frequency band is largely implicated in sleep, attention and decision-making processes (Nácher et al., 2013), while the medial prefrontal cortex is associated with decision-making and memory retrieval processes (Euston et al., 2012). Therefore, this component suggests that children with increased decision-making abilities may also have an increased ability to perform well in math related tasks. This hypothesis is consistent with long-standing evidence suggesting mathematics ability is associated with executive functioning and cognitive control in early childhood (Bull et al., 2011; Mazzocco & Kover, 2007).

The final component identified was a left hemisphere theta (4–8 Hz) component. This was predictive of global math ability (but not Applied Problems) and included BA 40 and various BAs in the superior region of the temporal lobe. This component is likely related to the attenuation of working memory functions as theta is related to concentration and the left temporal lobe is associated with receptive and expressive language as well as the phonological rehearsal processes implicit in working memory systems (Deschamps et al., 2020; Milner, 1968; Zulkurnaini et al., 2012). The lack of association between this component and the Applied Problems subtest may be explained given relations between the left temporal lobe and phonological processes used in working memory. The Applied Problem subtest involves listening to verbal math problems, thus intruding on one's ability to engage in the phonological rehearsal strategies that are crucial for some children to remember and perform the steps to complete basic math problems (Swanson, 2004; Swanson & Kim, 2007).

Overall, results demonstrate that in the right hemisphere increased alpha 1 coherence between BA 40 and the temporal lobe and reduced theta coherence between BA 40 and the frontoparietal region are associated with increased math abilities in children. Likewise, in the left hemisphere, increased delta coherence between BA 40 and the medial prefrontal cortex and increased theta coherence between BA 40 and the superior temporal lobe is associated with increased math abilities in children.

Notably, many of the components for predicting math abilities discovered in this study were also predictive of reading, except for a theta coherence network linking BA 40 to other BAs in the superior region of the temporal lobe. These results lend support to the domain-specific and domain-general hypotheses that surround neuroscientific research on learning disabilities (see Jäncke et al., 2019). Specifically, components related to deficits can be classified as domain-general, indicating that individuals with any type of learning disability can have underactive network connectivity in specified areas. However, some networks may represent specific deficits only present in individuals with a specific type of disability (i.e., math, reading, writing). The noted theta coherence component incorporating the left parietal and temporal lobes presents as a specific math-related network that is associated with diminished coherence in children with lower mathematics abilities.

When exploring the data further, a model containing three of the components (excluding the right alpha component) significantly predicts Broad Math scores, but not Broad Reading. This is suggestive of a cohesive domain-specific model of brain activity that underlies mathematical abilities more exclusively. Future studies should compare identified connectivity networks for both reading and mathematics ability to determine whether these networks are representative of specific or general academic abilities.

Future directions

Future studies may wish to utilize a larger sample size to examine cognition and math ability. Due to the relatively small sample size used in the current study, several steps were taken to reduce the number of variables within the dataset. The major consequences of this are two-fold; the first consequence results from performing PCA, which was performed to combine variables with redundancy. By combining multiple coherence variables into one component variable, spatial resolution of the anatomical structures involved in the coherence variables was sacrificed and the interpretability of brain regions represented by the variables was limited only to general brain regions rather than distinct BAs or brain structures. With a larger sample size, PCA may not be necessary since the large sample would not necessitate the reduction of variables, thus allowing researchers to examine coherence variables individually rather than in clusters of components. Though beyond the scope of the current study, eliminating PCA from the methods used would allow for the examination of variables that are related to specific math skills rather than global math ability. Furthermore, a large sample would allow additional analyses to examine differences between high achieving and low achieving mathematics students. Interestingly, interhemispheric coherence may be an especially important area to research as individuals who are gifted in math exhibit heightened interhemispheric connectivity compared to individuals who do not have exceptional math skills (O'Boyle, 2005).

Future neuroimaging research should aim to identify the function of the four coherence networks outlined in the current study. This can be achieved through functional neuroimaging techniques such as ERP, fNIRS, fMRI, MEG, etc. Since prior research suggests that mathematics skills rely on several key cognitive abilities, such as planning skills, attention, processing speed, working memory, and number sense (Van Luit & Toll, 2018), future studies may also aim to empirically determine which aspects of cognition each of the identified coherence networks are implicated in.

Lastly, more translational research is needed to increase the practical utility of qEEG research. To expand the practical utility of the current study's findings, future studies should pursue methods to address neural connectivity patterns that are consistent with poor academic performance; interventions such as neurofeedback therapy and applied behavior analysis have been used to address disconnection syndromes (e.g., Schummer, 2008) and should therefore be investigated as potential interventions aiming to mitigate the symptoms of MLD.

Conclusion

The current study is the first qEEG study to highlight empirical associations between BA 40 coherence and math ability. While the results presented in the current study offer novel information regarding cognition and math performance, their interpretation relies upon the assumption that a qEEG coherence network at rest correlating with a math score is indicative of a network that is directly involved in the performance of that math task. It is important to point out that, in fact, the four networks identified in the current study represent basal electrophysiological activity at rest and this resting-state brain activity does not imply that these networks are involved with the math tasks they are correlated with, nor that they become active during the performance of them. Despite this, the networks identified herein *are* predictive of math performance and therefore have utility in furthering the MLD knowledge-base, though interpretations warrant caution until further research validates these findings.

Despite this limitation, utilizing the method of EEG implementation and analysis outlined in this study can improve the understanding of how a developing brain performs math. By determining the degree of brain activity synchrony between specified locations throughout the brain at rest we can examine how these measures relate to broad academic skills and abilities. Our study provides a foundation of knowledge and unique research methodology upon which future studies examining the neurocognitive processes associated with academic ability may build upon. Additionally, we provide theoretical interpretations of the coherence networks with regards to how they relate to the underlying cognitive skills that are implicated in broad math ability. This information offers practical utility for MLD assessment and remediation by offering suggestions upon which interventions may target. For example, rather than focusing interventions for a student with MLD toward building basic math skills (e.g., addition and subtraction), our results provide additional support for interventions that target improving concentration and/or memory abilities along with basic math skills (e.g., Kroesbergen & Van Luit, 2003; Naglieri & Gottling, 1997).

Developing a research base within the areas of study that combine functional neurocognitive data and academic performance in children is lacking, but paramount if we hope to gain a better understanding of MLDs at a cognitive level. Exploring the neurocognitive underpinnings of academic performance is a major dearth in current neuropsychology and school psychology research. Investigation and research incorporating these practices lends promise for providing much needed improvements to our current learning disability assessment, identification, intervention, and remediation practices. The current study implicates general brain regions of interest for math ability, which include frontal, temporal and parietal regions in both hemispheres. The activity associated with these regions is related to specific cognitive abilities (i.e., concentration, attention, memory) that may also be important targets for math intervention, alongside typical academic interventions. Due to the exploratory nature of this research, findings warrant continued examination, yet provide a foundation for understanding some of the

neurological and cognitive underpinnings of math ability and/or performance in children.

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